

**AMENDMENT NO. 1 AUGUST 1996
TO
IS 12980 : 1990 FIREWORKS ROCKETS —
SPECIFICATION**

(Page 1, clause 2.1) — Substitute the following for the existing text:

The Indian Standards listed below are the necessary adjuncts to this standard:

<i>IS No.</i>	<i>Title</i>
301 : 1982	Potassium nitrate for explosive and pyrotechnic compositions (<i>second revision</i>)
1070 : 1992	Reagent grade water (<i>third revision</i>)
4905 : 1968	Methods for random sampling

(Page 1, clause 5, Title) — Substitute '**SIZES**' for '**GRADES**'.

(Page 1, clause **5.1**) — Substitute the following for the existing text:

'The fireworks rockets shall be of five sizes according to the requirement given in **6.1.6**.'

(Page 1, clause **6.1**) — Substitute the following for the existing text and renumber the subsequent clauses accordingly:

'6.1 Material and Construction

6.1.1 Composition — The chemical composition of pyrotechnic charge of fireworks rockets shall be as approved by the Chief Controller of Explosives.

6.1.2 Chlorate Content — The pyrotechnic composition, both in shell as well as in display component, shall be found to contain no chlorate when tested by the method prescribed in Annex A.

6.1.3 The Shell — The shell shall be made of kraft paper or paperboard, housing the propellant charge and the display component, and shall be of cylindrical shape all through its length. A nozzle shall be provided by means of either throttling the tube or by means of pierced hole through hardness earth or saw dust mixture. The diameter of the nozzle shall be such as to provide a good velocity for the rocket to push off vertically. The main shell shall have an empty extension below the nozzle in order to provide the propulsion chamber (*see* Fig. 1 and 2).

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6.1.3.1 The shell, when pressed hard at any place by means of thumb and index finger, shall not be deformed.

6.1.3.2 The shell shall not be made of any metal.

6.1.4 *The Stabilizer* — The stabilizer shall be straight, smooth, unbroken and uncracked all through its entire length. The length of the stabilizer shall be at least 5.5 times the height of the shell.

6.1.5 *The Fuse* — In all cases, the length of the quick match (fixed firmly through the nozzle by means of a priming composition) shall be not less than 30 mm outside the nozzle.

6.1.6 The nominal length and diameter of the shell, mass of pyrotechnic mixture and height of flight of the fireworks rockets shall be as follows

Size	Length of the Shell	Diameter of the Shell	Mass of Chemical Composition, Min	Height of Flight Min
	cm	cm	g	m
1	7.5 ± 0.1	1.5 ± 0.1	7.0	20.0
2	5.5 ± 0.1	1.5 ± 0.1	3.5	16.0
3	4.5 ± 0.1	1.5 ± 0.1	3.0	15.0
4	3.5 ± 0.1	1.5 ± 0.1	1.5	15.0
5*	7.5 ± 0.1	1.5 ± 0.1	8.0	40.0

*For whistling fireworks rockets only

6.1.7 The centre of gravity of the fireworks rocket shall be at a point 10 ± 1 mm below the lower edge of the propulsion chamber, when tested by the method prescribed in Annex B.

6.1.8 *Damp Proofness* — Fireworks rockets shall pass the test for damp proofness when tested by the method prescribed in Annex C.

[Page 3, clause 7.2(a)] — Substitute 'size' for 'grade'.

(Page 3, Annex A, Reference) — Substitute 'Clause 6.1.2' for 'Clause 6.1.1'.

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(*Page 4, Annex B, Reference*) — Substitute '*Clause 6.1.7*' for '*Clause 6.1.6*'.

(*Page 4, Annex C, Reference*) — Substitute '*Clause 6.1.8*' for '*Clause 6.1.7*'.

(CHD 026)